CS2320: Data Structures & Algorithms
Tutorial #03: Recursion

Part I: ***Choose the correct answer(s) and write the answers down in the table provided above:***

1. A method that calls itself is a _____ method.

a) Invalid
b) Static
c) Final
d) Recursive

Explanation: A recursive method is a method that calls itself.

2. _____ recursion occurs when a method calls itself, while _____ recursion when a method calls another method that then calls the original method.

a) upward, downward
b) downward, upward
c) direct, indirect
d) indirect, direct

Explanation: Direct recursion occurs when a method calls itself, while indirect recursion occurs when a method calls another method that then calls the original method.

3. _____ recursion results from the lack of a base case.

a) Indirect
b) Direct
c) Infinite
d) Spiral

Explanation: Infinite recursion results from the lack of a base case, which causes the recursive path to be followed forever.

4. The _____ puzzle is a favorite of computer scientists because of its elegant recursive solution.

a) Tower of Hanoi.
b) Sudoku.
c) Tetris.
d) Tic-Tac-Toe.

Explanation: The Tower of Hanoi puzzle has an elegant recursive solution, and has therefore become a favorite of computer scientists.

5. In the Towers of Hanoi puzzle, there are _____ pegs.

a) 2.
b) 3.
c) 4.
d) 5.

Explanation: In the Towers of Hanoi puzzle, there are always exactly three pegs.

6. In the Towers of Hanoi puzzle there are _____ disks of different diameters.

a) 3.
b) 4.
c) 5.
d) The Towers of Hanoi puzzle can include any number of disks of different diameters.

Explanation: There can be any number of disks in the Towers of Hanoi puzzle.

7. Which of the following statements is true?

a) Recursion should be used in every program.
b) Recursion should be avoided all the time.
c) Solutions that are easily expressed recursively should be programmed recursively.
d) Solutions that are easily expressed recursively should be programmed iteratively.

Explanation: Recursion should be used when the solution is easily expressed recursively.

8. Which of the following will usually result from infinite recursion?

a) The program will hang as though there is an infinite loop.

b) The program will not compile.

c) The program will run out of memory.

d) None of the above.

Explanation: Infinite recursion will cause a program to run out of memory.

9. In the Towers of Hanoi puzzle _____

a) It is legal to move a disk c isks.

b) It is legal to sit a disk asid

c) It is illegal to move a disk onto a peg that contains only larger disks.

d) None of the above are true

Explanation: Neither choice a, choice b or choice c make the statement true.

10. Almost all recursive programs _____.

a) Are more difficult to read than iterative programs.

b) Can be implemented iteratively.

c) Are easier to read than iterative programs.

d) Are shorter than iterative programs.

Explanation: Most recursive programs can be implemented iteratively, although a recursive solution may be easier to cod

11. The recursive solution of the Towers of Hanoi problem has _____ complexity.

a) Exponential.

b) Polynomial.

c) Logarithmic

d) None of the above

12. Which of the following is a proper recursive definition of *x* raised to the *y* power?

a) $x^y = x * x^{y-1}$

b) $x^y = x * x^{y-2}$

c) $x^y = x * x^{y-1}$ for $y > 1$; $x^y = x$ for $y = 1$

d) $x^y = x * x^{y-1}$ for all x

Explanation: Choice c is correct because it has a properly defined base case.

14. A solution with exponential complexity is _____.

a) Efficient.

b) Inefficient.

c) Easy to implement.

d) Difficult to implement.

Explanation: A solution with exponential complexity is very inefficient.

15. In a recursive solution, _____ is(are) always necessary.

a) Short, efficient coe.

b) Several variables.

c) Numerous lines of code.

d) Base case.

Explanation: A base case is always necessary in a recursive solution to avoid infinite recursion.

16. A recursive function defining the factorial for n > 1 is, the function facty(n):

a) facty(n) = n * facty(n-1)
b) facty(n) = facty(n) = n * facty(1-n)
c) facty(n) =n-1 * facty(n-2)
d) None of the above.

Explanation: Choice a is correct because it has properly defined recursive case.

17. A recursive function defining the Fibonacci number n ≥ 2 , Fabio(n) is

a) Fabio(n) =Fabio(n-1) + Fabio(1-n)
b) Fabio(n) =Fabio(n-1) + Fabio(n-2)
c) Fabio(n) =Fabio(n-1) + Fabio(Fabio n-2))
d) None of the above.